Amel KACI

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SUMMARY

Master's student in Computer Vision & Intelligent Systems with hands-on experience in computer vision, deep learning, and full-stack software development through academic and personal projects. Proficient in **Python, PyTorch, OpenCV and C++**, with practical exposure to model training and image processing. Seeking a **6-month internship** (Computer Vision / Deep Learning) starting February 2027 to apply technical skills to real-world impactful engineering problems.

EDUCATION

Master's Degree in Computer Vision and Intelligent Systems *September 2025 - Present*

Paris Cité University, Paris, France

- Relevant coursework: **Deep Learning, Computer Vision, Pattern Recognition, Data Science, Signal Processing, 3D imaging, Linear Algebra & Statistics**
- Key skills developed: **PyTorch model training and evaluation, Vision Transformers (ViT, DINO), CNN architectures, classical computer vision (SIFT, homography, segmentation), object detection with YOLO**

Double Bachelor's Degree in Computer Science and Japanese Studies *September 2022 - June 2025*

Paris Cité University, Paris, France

PROJECTS

Weakly Supervised Learning for Tumor Detection in Whole Slide Images (WSI) *Jan 2026 - May 2026*

Research Project – SIP Team at LIPADE, Paris Cité University

- Built a complete pipeline to detect tumors in **high resolution medical** images without pixel-level annotations, from image tiling and filtering to classification using **DistilBERT.**
- Cut compute time by treating the WSI as a 1D sequence and achieved a recall score of **0.91** for tumors.

Distributed Event Management Web Application (Django + React) *Mar 2026 – Apr 2026*

Web Development - Paris Cité University

- Developed a full-stack event-management platform with role-based authentication and permissions using **Django REST Framework** and **React for the frontend**.
- Containerized the application with Docker using a **microservice architecture**.

Low-level Coin Detection & Classification *Feb 2026 - Mar 2026*

Image Analysis and Processing Project - Paris Cité University

- Developed a classical image-processing pipeline for object detection and classification, using **Hough transform**, **Watershed segmentation**, M**orphological operations** and **Otsu's thresholding**.
- Delivered a lightweight, interpretable low-level approach with no dependency on heavy models.

Information Retrieval Engine *Feb 2026 - Apr 2026*

Data Science Project - Paris Cité University

- Designed a hybrid search engine combining lexical (**TF-IDF, BM25**) and semantic (**embeddings**) retrieval.
- Benchmarked the system in a **Kaggle competition** in teams of 4 (ranked 7th out of 19 teams).

SKILLS

Languages**: Python, C++, JavaScript, SQL, LaTeX** | Frameworks & Libraries: **PyTorch, TensorFlow, OpenCV, Django, React, NumPy, Pandas, scikit-learn** | Tools & Platforms: **Docker, Kubernetes, Git/GitHub, Jupyter**

Languages Spoken: **French** (Native), **English** (Fluent), **Japanese** (Intermediate)